We could instead consider splitting the dynamics up and performing the integration using a splitting method. Such a splitting strategy is considered in [232] where one uses

$$\begin{bmatrix} \mathrm{d}\boldsymbol{q} \\ \mathrm{d}\boldsymbol{p} \\ \mathbf{0} \end{bmatrix} = \underbrace{\begin{bmatrix} \boldsymbol{M}^{-1}\boldsymbol{p}\mathrm{d}t \\ -\nabla U(\boldsymbol{q})\mathrm{d}t - \boldsymbol{g}'(\boldsymbol{q})^T\boldsymbol{\lambda}\,\mathrm{d}t \\ \boldsymbol{g}(\boldsymbol{q}) \end{bmatrix}}_{\mathrm{H}_c} + \underbrace{\begin{bmatrix} \mathbf{0} \\ -\gamma\boldsymbol{p}\mathrm{d}t + \sqrt{2\gamma k_{\mathrm{B}}\mathrm{T}\boldsymbol{M}}\mathrm{d}\mathbf{W} - \boldsymbol{g}'(\boldsymbol{q})^T\boldsymbol{\lambda}\,\mathrm{d}t \\ \boldsymbol{g}(\boldsymbol{q}) \end{bmatrix}}_{\mathrm{O}_c},$$

splitting the dynamics into two pieces: the constrained Hamiltonian (NVE) equations, and a constrained Ornstein-Uhlenbeck piece.

We solve these two pieces symmetrically, where the constrained Hamiltonian part is solved over a step h, in between two solves of the Ornstein-Uhlenbeck SDE with stepsize $h/2$. The SHAKE (or RATTLE) algorithm can be used to solve the constrained Hamiltonian part to a high degree of accuracy, while solving the Ornstein-Uhlenbeck piece along the manifold amounts to finding the solution of a linear system. We observe the constraint in both solutions, so that the computed points lie in the co-tangent bundle of the constraint manifold. Instead of solving the Ornstein-Uhlenbeck exactly, we discretize using a midpoint Euler scheme, yielding the following scheme:

Midpoint Euler - Verlet - Midpoint Euler (see [232])

$$\text{Solve} \begin{cases} \boldsymbol{p}_{n+1/4} = \boldsymbol{p}_n - \frac{h\gamma}{4}(\boldsymbol{p}_n + \boldsymbol{p}_{n+1/4}) + \sqrt{h\gamma k_{\mathrm{B}}\mathrm{T}\boldsymbol{M}}\boldsymbol{R}_n - \boldsymbol{g}'(\boldsymbol{q}_n)^T\boldsymbol{\lambda}_{n+1/4} \\ \text{Such that } \boldsymbol{g}'(\boldsymbol{q}_n)\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/4} = \mathbf{0} \end{cases}$$

$$\text{Solve} \begin{cases} \boldsymbol{p}_{n+1/2} = \boldsymbol{p}_{n+1/4} - \frac{h}{2}\nabla U(\boldsymbol{q}_n) - \boldsymbol{g}'(\boldsymbol{q}_n)^T\boldsymbol{\lambda}_{n+1/2} \\ \boldsymbol{q}_{n+1} = \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2} \\ \boldsymbol{p}_{n+3/4} = \boldsymbol{p}_{n+1/2} - \frac{h}{2}\nabla U(\boldsymbol{q}_{n+1}) - \boldsymbol{g}'(\boldsymbol{q}_{n+1})^T\boldsymbol{\lambda}_{n+3/4} \\ \text{Such that } \boldsymbol{g}'(\boldsymbol{q}_{n+1})\boldsymbol{M}^{-1}\boldsymbol{p}_{n+3/4} = \mathbf{0} \text{ and } \boldsymbol{g}(\boldsymbol{q}_{n+1}) = \mathbf{0}. \end{cases}$$

$$\text{Solve} \begin{cases} \boldsymbol{p}_{n+1} = \boldsymbol{p}_{n+3/4} - \frac{h\gamma}{4}(\boldsymbol{p}_{n+3/4} + \boldsymbol{p}_{n+1}) + \sqrt{h\gamma k_{\mathrm{B}}\mathrm{T}\boldsymbol{M}}\boldsymbol{R}_{n+1/2} - \boldsymbol{g}'(\boldsymbol{q}_{n+1})^T\boldsymbol{\lambda}_{n+1} \\ \text{Such that } \boldsymbol{g}'(\boldsymbol{q}_{n+1})\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1} = \mathbf{0} \end{cases}$$

The vectors $\boldsymbol{R}_n$ and $\boldsymbol{R}_{n+1/2}$ are vectors of $\mathcal{N}(0,1)$ i.i.d. random numbers, with $\gamma \geq 0$ the usual Langevin dynamics friction parameter. Setting $\gamma = 0$ reduces the scheme to the usual RATTLE scheme for solving holonomically constrained Hamiltonian equations of motion, whereas if γ is chosen large then this will be expected to cause instability in the scheme, as we are not solving the OU process exactly. The

Lagrange multiplier $\boldsymbol{\lambda}_{n+1/4}$ (and likewise $\boldsymbol{\lambda}_{n+1}$) can be obtained by solving the linear system

$$\boldsymbol{g}'(\boldsymbol{q}_n)\boldsymbol{M}^{-1}\left[\left(1-\frac{h\gamma}{4}\right)\boldsymbol{p}_n+\sqrt{h\gamma k_{\mathrm{B}}\mathrm{T}\boldsymbol{M}}\boldsymbol{R}_n-\boldsymbol{g}'(\boldsymbol{q}_n)^T\boldsymbol{\lambda}_{n+1/4}\right]=\mathbf{0},$$

which is easily computed once the $\boldsymbol{R}_n$ vector is evaluated in the step, and gives us a stability criterion

$$\gamma h < 4. \tag{7.46}$$

A similar calculation and linear solve is needed to compute $\boldsymbol{\lambda}_{n+3/4}$ in order to maintain the hidden constraint for $\boldsymbol{p}_{n+3/4}$, whereas the Lagrange multiplier $\boldsymbol{\lambda}_{n+1/2}$ is responsible for ensuring that $\boldsymbol{g}(\boldsymbol{q}_{n+1})=\mathbf{0}$, and is typically found using an implicit solve.

There are many other ways we could choose to combine traditional molecular dynamics discretization schemes with constrained solvers such as variants of the SHAKE algorithm (see, for example [81, 383, 396]). The following section presents an alternative based upon the work on splitting methods in this chapter so far.

7.12.1 A "Geodesic Integrator"

We may consider alternative splittings of the dynamics in order to build a library of available schemes, just as we did in the unconstrained case in Sect. 7.3.1. Solving the constrained OU part exactly would allieviate the condition (7.46), while we may hope to reduce discretization error in the spirit of the success of the ⟦BAOAB⟧ scheme in Sect. 7.9.3. The natural splitting template to consider would be to again split into potential and kinetic energies, with the constrained Ornstein-Uhlenbeck piece:

$$\begin{bmatrix}\mathrm{d}\boldsymbol{q}\\ \mathrm{d}\boldsymbol{p}\\ \mathbf{0}\end{bmatrix}=\underbrace{\begin{bmatrix}\boldsymbol{M}^{-1}\boldsymbol{p}\,\mathrm{d}t\\ -\boldsymbol{g}'(\boldsymbol{q})^T\boldsymbol{\lambda}\,\mathrm{d}t\\ \boldsymbol{g}(\boldsymbol{q})\end{bmatrix}}_{\mathrm{A}_c}+\underbrace{\begin{bmatrix}\mathbf{0}\\ -\nabla U(\boldsymbol{q})\mathrm{d}t-\boldsymbol{g}'(\boldsymbol{q})^T\boldsymbol{\lambda}\,\mathrm{d}t\\ \boldsymbol{g}(\boldsymbol{q})\end{bmatrix}}_{\mathrm{B}_c}$$
$$+\underbrace{\begin{bmatrix}\mathbf{0}\\ -\gamma\boldsymbol{p}\mathrm{d}t+\sqrt{2\gamma k_{\mathrm{B}}\mathrm{T}\boldsymbol{M}}\mathrm{d}\mathbf{W}-\boldsymbol{g}'(\boldsymbol{q})^T\boldsymbol{\lambda}\,\mathrm{d}t\\ \boldsymbol{g}(\boldsymbol{q})\end{bmatrix}}_{\mathrm{O}_c}.$$

Observe that the holonomic constraint (as well as the hidden constraint) is maintained in each part, changing the usual A piece into a *geodesic drift* along the